Rafayet Hossain

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Education

Hunter College, City University of New York (CUNY) Expected May 2027
Bachelor of Arts in Computer Science, Minor in Mathematics | GPA: 3.61/4.00 *New York, NY*

- Relevant Coursework: Software Engineering, Computer Architecture, Operating Systems, Network Security, Database Management, Automation Engineering, Discrete Math, Modern AI Systems
- Honors & Awards: Dean's List, Peter F. Vallone Academic Scholarship

Skills

- **Programming:** C++, Python, C, Bash, Node.js, Dart, TypeScript, JavaScript, HTML, CSS, Markdown
- **Technologies:** Git, GitHub, Linux, Docker, GitHub Actions, TimescaleDB, PostgreSQL, SQL Server, Make, GCC, KVM, Wireshark
- **Libraries/Frameworks:** Next.js, React, Flutter, Supabase, Firebase, TailwindCSS, PyTorch,OpenCV, Jekyll, Pygame, Pyinstaller
- **Certifications:** Codepath Applications of AI Engineering (Aug 2026), Codepath Intermediate Cybersecurity (Apr 2024)

Experience

Research Foundation of CUNY Aug 2025 — Present
Systems Engineering Intern *New York, NY*

- Diagnosed a SQL Server pipeline failure in automated document ingestion, tracing SQL Server Agent jobs and migration logs to code errors introduced by a server migration.
- Implemented automated data retention and backups for 20+ employee workstations at a New York City government office.
- Developing a software extension for energy imputation research at CUNY Queens College buildings, evaluating machine learning methods such as MICE and BRITS to fill missing data.

CUNY Tech Prep Aug 2025 — May 2026
Fellow, Web Development *New York, NY*

- Selected for a competitive web development fellowship with students from across the 11 CUNY senior colleges, where Fellows create technical projects using tools such as React, Node + Express, and PostgreSQL.
- Learned industry best practices for design, implementation, and deployment, such as MVC, version control with Git/GitHub, agile & Scrum with Trello and Slack, test-driven development, and CI/CD.

Jupiter Leo Productions Jul 2025 — Aug 2025
Software Development Intern *Remote*

- Built responsive and accessible educational games with a team of 3 using HTML, CSS, and JavaScript, improving course engagement and learning outcomes by 34%.
- Enhanced a financial education platform with media and interactive elements, improving the experience for 100+ users.
- Integrated generative AI tools into content creation, reducing manual design effort and production time by 50%.

Projects

Flocksync | *Flutter, Firebase, Firestore, Dart, Node.js, OpenStreetMap* *View Source*

- Built a cross-platform rental management application in Flutter to help tenants and property managers centralize communication, maintenance, and property-related workflows.
- Implemented Firebase Authentication and Firestore data models for secure account creation, tenant onboarding, role-based access control, and real-time app updates.
- Integrated OpenStreetMap APIs for address autocomplete and verification through a Node.js backend, significantly improving property lookup and input accuracy.

Zig Actions Runner | *Zig, Dagger, GitHub Actions, TimescaleDB, PostgreSQL, React, TypeScript* *View Site*

- Won Best Use of Tiger Data at the AI Engineer NYC 2026 Hackathon for building a lightweight, self-hosted GitHub Actions runner in Zig that runs workflow steps as cached Dagger pipelines, reducing repeated job times by 80%.
- Designed the Tiger Data (TimescaleDB) schema for runs, jobs, and logs, using time-based partitioning, compression, and data retention policies to store high-volume log data efficiently.
- Built a React dashboard in TypeScript that shows each runner's status and its running and finished jobs, with automated tests.

HerRide | *Next.js, Node.js, Supabase, Clerk, TypeScript, TailwindCSS* *View Source*

- Developed a full-stack, women-only ridesharing platform using Next.js in a team of 4 as part of CUNY Tech Prep.
- Integrated secure user authentication and management using Clerk and designed PostgreSQL schemas with role-based access using Supabase.
- Contributed to product design and planning by prototyping in Figma and coordinating features, requirements, and tech stack decisions using Trello and Notion.